RxGuard: A COMPUTER VISION-BASED ELECTRONIC HEALTH RECORD SYSTEM FOR MEDICAL RECORD DIGITIZATION WITH IMAGE DEDUPLICATION

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

Reliable data is the backbone of any effective healthcare system. Today, the healthcare industry is transitioning away from physical paper charts toward Electronic Health Records (EHRs) because centralized data enables doctors to make better, faster decisions [1] [2]. It is not just about going paperless; it is about ensuring that a patient's medical history—their allergies, past surgeries, and lab results—is available exactly when it is needed. In Southeast Asia, research indicates a significant push toward digital health solutions to enhance how hospitals manage information and expand their reach to more patients [3]. The goal is straightforward: to ensure that care continues smoothly even if a patient transitions from one clinic to another.

However, moving to digital systems is easier said than done. Even when hospitals invest in expensive software, they often encounter "post-implementation challenges" where the new system fails to integrate seamlessly with the daily workflow of nurses and doctors [4]. A major problem is that many of these systems operate in isolation; different platforms cannot share patient records with one another. This creates "data silos," where a patient's information is confined to one location and cannot be accessed by a doctor in another hospital. This lack of connection is dangerous. When pharmacists or doctors cannot see the full picture, they might miss a drug interaction or duplicate a test. In fact, medication errors—often caused by incomplete information—remain a leading cause of preventable harm in healthcare [1].

The danger of incomplete information is even more alarming when viewed through the lens of the current cybersecurity landscape in the Philippines. In 2023, the healthcare sector faced a massive crisis during the PhilHealth Medusa ransomware

attack, where the personal and medical data of over 13 million unique members were compromised and leaked online due to a lack of adequate protection like antivirus software [1]. This incident is a clear indication that "Information Security" in local health institutions is still at a critical stage, as traditional safeguards like simple passwords or basic firewalls can no longer withstand modern hacking techniques used by ransomware groups [2]. According to the National Privacy Commission (NPC), the healthcare industry remains one of the top targets for data breaches in the country, often leading to the sensitive records of Filipinos being exposed to unauthorized parties.

This lack of security also opens the door to falsification, which is a major problem in paper-based or poorly managed digital systems. Without a strict verification process, physical prescriptions and medical results are highly prone to being tampered with or forged to obtain regulated medications or to misrepresent health insurance claims. Studies comparing electronic versus paper-based systems confirm that reliance on physical paper significantly increases errors and lacks the security found in digital prescribing [3], [4]. Since physical papers do not have a digital audit trail, it becomes nearly impossible to track who altered the data, putting both the doctor's license and the patient's safety at risk.

To solve these systemic issues, the industry is looking toward emerging technology, particularly the use of Artificial Intelligence (AI) to automate and secure data handling. This is where Computer Vision comes in as a primary solution. Recent research demonstrates that Computer Vision can effectively digitize paper health records using smartphone images, even in low-resource settings like the Philippines [5]. This field of AI trains computers to interpret and understand the visual world—essentially giving the system the ability to "see" and process documents like a human would, but with much higher accuracy.

Computer Vision serves as the best solution for RxGuard because it removes the need for manual encoding, which is identified as a primary source of data inaccuracy [6], [7]. By using algorithms to scan and recognize text from physical records, the system ensures that the digitized version is a verified copy. Furthermore, advanced Deep Learning approaches can now post-correct OCR results to ensure high accuracy in medical reports [8]. This technology doesn't just digitize files; it acts as a security layer that prevents falsification by ensuring that every record uploaded is unique, authentic, and protected by a digital trail that manual paper filing can never provide.

In the Philippines, the situation is even more complicated. While there have been pioneering efforts, such as the Community Health Information and Tracking System (CHITS) for local health centers [5], the overall landscape remains highly fragmented. Studies on Philippine health information systems show that most public and private facilities operate independently [6]. This forces Filipino patients to act as their own "record keepers." They often have to carry brown envelopes filled with old prescriptions and lab results from doctor to doctor. If these papers get lost—whether due to flooding, misplaced files, or simple wear and tear—the medical history is gone. Formative evaluations of eHealth in the country highlight that poor infrastructure and lack of resources make it hard to maintain consistent digital records [7].

In addition to the logistical nightmare, there is the issue of trust and security. Medical records contain sensitive personal details, and protecting them is required by law under the Data Privacy Act of 2012 (RA 10173) [8]. However, compliance is a struggle for many local institutions due to a lack of technical expertise and budget [9]. The risks are real. We have seen high-profile data breaches in the news, such as incidents involving psychiatric or sensitive health records, which expose the vulnerabilities of current systems [10]. Traditional security measures, such as simple passwords, are no longer sufficient to stop modern cyber threats [11]. If patients feel their data is not safe, they will be hesitant to support digital transformations.

A specific technical gap also remains: how do we get the data from the paper into the computer. Currently, most clinics rely on manual encoding, where staff manually type details. This is slow, boring, and prone to human error. A tired staff member might type the wrong dosage or misspell a patient's name, resulting in discrepancies between the prescribed medication and the information in the system. Also, without a smart way to check files, a system can get clogged with duplicate images or records, making it hard to know which one is the "true" copy.

This is where RxGuard comes in. The proposed system is designed to be a Computer Vision-Based Electronic Health Record System for Medical Record Digitization with Image Deduplication platform that solves these specific problems. Instead of relying solely on manual typing, RxGuard utilizes Computer-Driven Document Digitization and Optical Character Recognition (OCR) to automatically read and convert physical medical documents into digital data. It also features Image Deduplication to ensure that every record uploaded is unique and verified. By bridging the gap between the physical pile of papers and a secure digital database, RxGuard aims to provide a solution that is not only secure and compliant with Philippine laws but also practical for the real-world needs of doctors and patients.

1.2 Objectives

The general objective of this study is to develop RxGuard: Computer Vision-Based Electronic Health Record System for Medical Record Digitization with Image Deduplication, designed to facilitate controlled access to patient data while aligning with data privacy and cybersecurity standards.

The specific objectives of the study are listed below:

- To design a computer vision framework capable of accurately scanning and digitizing handwritten and printed medical records.

- To integrate Optical Character Recognition (OCR) algorithms for the automated processing and digitization of physical medical documents.
- To implement an image deduplication algorithm that detects and removes duplicate medical record images.
- To evaluate the accuracy and reliability of the system in comparison to traditional medical record management methods.

1.3 Significance of the Study

This study is significant because it develops a centralized system, RxGuard, that directly tackles the critical gap in EHR verification between doctors and pharmacists. By developing RxGuard, a computer-vision based Electronic Health Record (EHR) platform with automated medical record digitization and image deduplication, the study provides a practical, technology-driven solution that improves healthcare data management while adhering to data privacy and cybersecurity standards.

The results of the study will bring benefits to the following:

Patients. This system empowers patients by formally linking the visibility of their medical records to the consent they have provided. It protects their privacy by ensuring that sensitive information is not freely accessible to all users, such as pharmacists, without their agreement. This control can strengthen trust in the healthcare system and increase a patient's confidence that their data is being handled responsibly, potentially leading to better engagement with their treatment plans.

Doctors and Pharmacists. These primary users benefit from a streamlined and safer workflow. Doctors can register patients and issue prescriptions through a unified system, reducing administrative time. Pharmacists gain a crucial safety net, allowing them to verify prescriptions for patients who have lost their physical copies, directly

addressing a common point of failure. This enhances clinical safety, reduces medication errors, and improves inter-professional communication.

Healthcare Administrators. Administrators gain powerful oversight tools through the system's centralized dashboard. They can monitor key operational metrics in real-time, such as the number of registered entities (hospitals, pharmacies, doctors) and patient registrations. This data facilitates resource allocation, simplifies system management, and enables proactive monitoring of adoption rates and system-wide activity, ultimately leading to more efficient governance.

Developers and Researchers. This project serves as a practical implementation of an advanced security model, Contextual Role-Based Access Control (C-RBAC), in a healthcare context. It demonstrates how to build a system that dynamically adjusts user permissions based on context, like consent status. This can inspire future research and development in secure, privacy-compliant health information systems, providing a valuable case study for capstone projects in IT and health informatics.

1.4 Scope and Limitations

The primary focus of this study is the development of RxGuard, a web-based Electronic Health Record (EHR) platform designed to digitize, secure, and manage patient medical records. The system utilizes Optical Character Recognition (OCR) to convert physical documents into digital text and implements Image Deduplication to ensure that no duplicate records exist in the database.

The functional scope is divided into four main user roles:

1. Administrator

   The Admin serves as the system overseer. Their primary function is to manage the registration of hospitals, clinics, doctors, and pharmacists, ensuring that only verified entities can access the system. They have access to a

dashboard that shows the total number of users and system activities. Crucially, the Admin is responsible for reviewing and approving patient consent forms uploaded by doctors, serving as the final checker before a patient's record becomes viewable by pharmacists.

2. Doctor

Doctors play a central role in data entry. Their scope includes:

- Patient Registration: Creating accounts for patients and uploading their signed consent forms.
- Document Digitization: Uploading images of medical records (such as lab results). The system automatically scans these images using OCR to extract text, reducing the need for manual typing.
- Record Management: Viewing and updating the medical history of patients under their care.

3. Pharmacist

The scope for pharmacists is strictly limited to viewing and verification. They can search for a patient using a unique ID to view their prescription history. However, they can only view these records if the patient has granted active consent. This module is designed to help pharmacists verify if a prescription is authentic before dispensing medication.

4. Patient

Patients are empowered to manage their data privacy. Their scope allows them to grant or revoke permission for pharmacists to view their records through a consent form. This ensures that they remain the owners of their personal health information.

Limitations of the Study

To ensure the feasibility of the project within the given timeframe, the following limitations are set:

- No Clinical Decision Support (CDS): The system is strictly a record management platform. It does not provide medical advice, diagnosis, or automatic alerts for drug medications (e.g., warning if Patient is allergic to Medicine X). The doctor remains solely responsible for all medical decisions.
- OCR Dependency on Document Quality: The accuracy of the Computer-Driven Medical Record Digitization (OCR) feature depends heavily on the legibility of the physical document. While the system can read printed text and clear handwriting, it may struggle with highly cursive or illegible handwriting, which is common in medical prescriptions. In such cases, manual correction by the doctor may still be required.

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter describes the studies that the system's development and related existing systems have in common. These served as the researchers' knowledge base when building the plugin, as well as helping them anticipate issues that would arise later in the development process.

2.1 Related Literature

2.1.1 EHR Adoption and Digital Health Transformation in the Philippines

The Philippines' journey toward adopting Electronic Health Record (EHR) systems reflects a broader regional trend of digital health innovation prompted by both policy imperatives and practical healthcare challenges [9]. Systems like the Community Health Information and Tracking System (CHITS) and OpenMRS have become pillars of this evolution, providing adaptable platforms for local government units (LGUs) and rural clinics to digitize health records, enhance data management, and streamline service delivery [10] The adoption of EHRs is partly driven by national goals for universal healthcare coverage and improved health information management, but the journey remains uneven, especially in public sector settings, where resource disparities, technical hurdles, and limited internet infrastructure frequently hinder implementation and sustainability [11] [12] [13].

Multiple studies emphasize that the post-implementation phase is critical yet often neglected. After initial system deployment, organizations struggle to ensure that EHRs deliver improved outcomes, integrate smoothly with existing clinical and administrative workflows, and remain technically robust in the face of growing user demands and evolving regulations [11] [14].

Hardware and technical limitations are major barriers to effective EHR use, as are challenges related to inconsistent ICT infrastructure and staff training, particularly in remote or underserved areas [12] [13]. These barriers result in significant differences between public and private healthcare institutions: public sector providers often encounter more severe infrastructure deficits and bureaucratic roadblocks, while private providers, although relatively better resourced, must still contend with high implementation costs, interoperability issues, and regulatory compliance requirements [12] [13].

The perceived benefits of EHR adoption include increased clinical process efficiency, better data quality, and more consistent collaboration among healthcare team members [15]. However, end-user acceptance is typically contingent upon system usability, workflow relevance, and the degree of co-design with healthcare staff [14]. Empirical evidence from the Philippines and comparable low- and middle-income countries (LMICs) indicates that the success of digital health transformations is enhanced when healthcare workers, especially nurses and physicians, are engaged in design and rollout, and when targeted investments in digital literacy and ongoing technical support accompany EHR deployment [9] [15].

2.1.2 Privacy and Cybersecurity Challenges in EHR Systems

The increasing digitization of health records has raised significant concerns about data privacy, cybersecurity, and patient trust [16]. The Philippines has not been immune to high-profile healthcare data breaches, such as the PhilHealth incident, which dramatically underscored the risks posed by inadequate security protocols and fragmented data governance [17] [18]. The Philippines has not been immune to high-profile healthcare data breaches, such as the PhilHealth incident, which dramatically underscored the risks posed by inadequate security protocols and fragmented data governance [18].

Traditional cybersecurity measures, such as password protection, basic encryption, and firewalls, now prove insufficient against sophisticated threats, including ransomware, insider access, and phishing scams [19] [20]. Accordingly, studies increasingly recommend the integration of dynamic, human-centric threat detection and mitigation, leveraging artificial intelligence (AI) and machine learning to rapidly spot and respond to anomalies [20] [21].

International frameworks, such as the World Health Organization's Global Strategy on Digital Health 2020–2025, stress the importance of harmonized data standards, comprehensive risk assessment, and continuous policy evolution to keep pace with both technological innovation and new classes of cyber threats [22]. Within the Philippines, the Data Privacy Act of 2012 (RA 10173) serves as the legal foundation for data protection; however, full compliance is rare, constrained by technical, financial, and organizational factors. Research indicates that many Filipino healthcare organizations remain only partially compliant, hindered by low awareness, insufficient resources, and fragmented governance structures [23] [24]. The National Privacy Commission (NPC) plays a crucial enforcement and advisory role; however, the actual penetration of robust privacy practices is uneven, especially outside major urban centers [25].

2.1.3 Consent-Based and Role-Based Access Control in Healthcare Information Systems

Granular access control models are top priorities in modern EHRs, enabling organizations to segment sensitive data and ensure that only authorized users can retrieve vital information [26]. Role-Based Access Control (RBAC), the most widely implemented approach, restricts access to prescription and clinical data based on user roles (e.g., pharmacist, clinician) and has proven especially effective when carefully tailored to healthcare

workflows. However, emerging studies highlight limitations of RBAC, for instance, its inability to factor in dynamic patient-provider relationships or explicitly honor individual patient consent [27]. To address this, Relationship-Based Access Control (ReBAC) models and consent-based frameworks are actively explored in both scholarly and applied settings, enabling systems to enforce "need to know" principles based on contextual patient-provider interactions and explicit patient permissions [27] [28] [29].

Recent advances leverage blockchain technology and smart contracts to automate consent management, providing transparent, tamper-evident, and patient-driven control over data access. These approaches empower patients to decide precisely when, for how long, and to whom their records may be disclosed, bolstering both legal compliance and patient trust. Technical evaluations reveal that performance bottlenecks, such as transaction throughput and scalability, can be mitigated by utilizing hybrid architectures, including off-chain decentralized storage solutions for large datasets [30].

Consent management is further reinforced by cryptographic primitives such as conditional proxy re-encryption, which allows encrypted data to be securely re-shared upon patient approval without exposing it to system administrators or unauthorized third parties [29]. Such approaches are increasingly recommended in policy guidance and regulatory compliance models, notably in the EU's GDPR context and by the Philippines' NPC [25] [31].

2.1.4 The Evolving Role of Pharmacists and Clinical Integration

Digital health transformation has fundamentally altered the professional landscape of pharmacists, increasingly positioning them as pivotal collaborators within care teams via EHR integration. In environments where

pharmacists have structured access to relevant clinical data, studies have documented significant improvements in medication therapy management, patient safety, and medication adherence, particularly for the management of chronic diseases such as diabetes, hypertension, and asthma. Integration of pharmacists into EHR workflows reduces delays, eliminates redundant communication, and ensures timely identification of potential drug interactions and contraindications, thus preventing adverse events [32] [33].

Nevertheless, the literature highlights ongoing barriers, including fragmented systems, inconsistent or insufficient training, and persistent technical usability challenges, which can undercut the intended benefits [34] [38]. The demand for pharmacist access to clinical data, supported by modular, consent-aware plug-ins in open-source EHRs, is well recognized, as is the need for robust audit logging and privacy controls to support regulatory compliance and professional accountability [32] [34]. International examples, including those from the US, Europe, and Australia, underscore the feasibility and clinical value of such integrations, propelling similar innovation in the Philippine context [33].

2.1.5 Optical Character Recognition (OCR) in Document Digitization

Optical Character Recognition (OCR) has been extensively studied as a fundamental technology for converting scanned or captured document images into machine-readable text. Zhang, Bai, and Cui conducted a comprehensive review of OCR systems based on artificial intelligence algorithms, emphasizing the transition from traditional machine learning techniques to deep learning-based approaches [34]. Their study highlighted that deep learning models significantly outperform earlier OCR methods in handling noisy images, varied fonts, and irregular character spacing. However, the authors also noted challenges related to computational cost and optimization

when deploying such models in real-world systems. These findings support the adoption of AI-based OCR techniques in document digitization platforms that process heterogeneous data, such as healthcare records.

Patil et al. investigated the use of semantic segmentation as a preprocessing step to enhance OCR accuracy in documents containing mixed text types, including handwritten and printed characters [35]. The study demonstrated that isolating text regions prior to OCR processing reduces background interference and improves recognition accuracy. This approach was shown to be particularly effective for documents with complex layouts, which are common in medical prescriptions and forms. The results emphasize the importance of robust preprocessing pipelines in OCR-driven systems.

A study published by Digital Innovation examined the performance of OCR engines, particularly Tesseract, in processing scanned documents [36]. The researchers observed that while Tesseract performs well on clean, printed text, its accuracy declines when dealing with low-quality images or handwritten content. The study concluded that preprocessing techniques such as binarization, noise reduction, and normalization are essential to improve OCR reliability. This finding reinforces the necessity of integrating OCR engines with image enhancement techniques in practical applications.

2.1.6 Practical OCR Tools and Systems

OCRopus is an open-source OCR framework designed to support modular document analysis pipelines [37]. Unlike traditional monolithic OCR systems, OCRopus allows customization of individual components such as layout analysis, text line detection, and character recognition. Its flexibility enables adaptation to domain-specific datasets and experimental

configurations, making it suitable for research-oriented document digitization systems.

Several applied studies have evaluated the effectiveness of the Tesseract OCR engine in real-world scenarios. Sharma and Gupta demonstrated that Tesseract achieves acceptable accuracy levels for printed documents when combined with appropriate preprocessing techniques [38]. Similarly, Kumar and Singh emphasized that Tesseract's open-source nature makes it cost-effective and extensible, but also highly dependent on image quality and language model training [39]. These studies indicate that while Tesseract is a viable OCR solution, its performance is significantly enhanced when integrated with preprocessing and post-processing mechanisms.

2.1.7 Image Deduplication in Computer Vision Systems

Image deduplication has become an important area of research in computer vision, particularly for systems managing large volumes of image data. Li et al. proposed CE-Dedup, a framework designed to remove near-duplicate images from datasets used to train convolutional neural networks (CNNs) [40]. Their experiments showed that eliminating redundant images reduced dataset size by up to 75% without negatively affecting model accuracy. The study concluded that duplicate images contribute minimal learning value while increasing computational cost, highlighting the importance of deduplication for efficiency and data integrity.

Shah and Shrivastava introduced a semantic-aware image deduplication approach that integrates object recognition with traditional visual similarity measures [41]. Unlike pixel-based or hashing-only methods, their approach captures high-level semantic features to identify duplicates more accurately under varying conditions such as changes in lighting, orientation, and image

quality. The results demonstrated improved duplicate detection accuracy, particularly for visually different images that convey the same content. This study underscores the importance of semantic understanding in advanced deduplication systems.

2.1.8 Computer Vision and Theoretical Foundations of OCR Systems

Computer vision serves as the fundamental framework that enables both optical character recognition (OCR) and image deduplication processes in modern document digitization systems. Litjens et al. presented a comprehensive survey on the application of deep learning in medical image analysis, emphasizing how convolutional neural networks extract hierarchical visual features from images [42]. These features support a wide range of downstream tasks, including text recognition, image classification, and image similarity comparison. Such capabilities form the technical backbone of contemporary OCR and deduplication workflows, justifying the integration of computer vision techniques in document processing systems.

The theoretical foundations of OCR systems have evolved alongside advancements in computer vision. A review published in the International Journal of Engineering Research & Technology traced the progression of OCR technologies from early template matching and rule-based methods to modern machine learning and deep learning approaches [43]. The authors outlined the core stages of OCR processing, including image preprocessing, feature extraction, character classification, and post-processing. Despite improvements in recognition accuracy through advanced algorithms, the study emphasized that OCR performance remains sensitive to image quality and document layout inconsistencies. This highlights the continued importance of preprocessing and validation mechanisms to ensure reliable OCR results in real-world applications.

2.1.9 Regulatory Landscape and Compliance with Data Privacy Act (RA 10173)

Philippine literature on health data privacy and regulatory compliance reveals persistent gaps between legislative mandates and real-world practice [36]. Many health sector organizations are only partially compliant due to a lack of resources, technical expertise, and a clear roadmap for systematic alignment with data protection law. Factors such as lack of awareness, budget constraints, a "wait-and-see" institutional attitude, and insufficient staffing frequently hamper compliance efforts [37] [32].

Operationalizing data privacy compliance entails not just legal conformity but practical integration into workflows, system design, and staff behavior. Recommended actions include capacity-building programs, Privacy Impact Assessments, and continuous policy review aligned with international best practices, such as the ISO/IEC 27001 standards [25] [34]. Importantly, compliance is not static: it demands periodic reassessment in light of evolving risks, new system deployments, and changing policy environments.

2.2 Related Systems

2.2.1 The Community Health Information and Tracking System (CHITS)

CHITS is the Philippines' first widely used open-source EHR system, tailor-made for use in local government unit (LGU) health centers and designed with participatory, user-driven methodologies. Notably, CHITS prioritizes the collection and management of primary care data, supports public health program monitoring, and has demonstrated measurable improvements in efficiency, data quality, and health worker morale. However, as digital healthcare requirements have evolved, particularly in terms of privacy and

pharmacist integration, CHITS's limitations have become more pronounced. Challenges include insufficient technical robustness, a lack of interoperability standards, and, notably, the absence of sophisticated, consent-driven external access controls necessary for secure data sharing between pharmacists and third parties [42][43]. The decentralized governance of Philippine healthcare further complicates CHITS implementation, as each LGU may pursue different priorities or investments in digital health [41]. The system's success continues to hinge on alignment with local policy and sustained user engagement, but meaningful data exchange and integration with advanced privacy features remain developmental priorities [39] [40].

2.2.2 OpenMRS: Modularity, Plug-Ins, and Pharmacy Integration

OpenMRS is an open-source EHR platform celebrated globally for its modular, extensible architecture, enabling tailored adaptation to diverse clinical settings and health system requirements. Implemented in various Philippine rural clinics, OpenMRS allows for the dynamic addition of plug-in modules, making it a strong candidate for incremental feature enhancements, such as pharmacist access, consent management, and audit logging. The introduction of advanced access control extensions, including Relationship-Based Access Control (ReBAC) and improved RBAC, permits fine-grained authorization policies that are sensitive to user-patient relationships and contextual permissions, building on and surpassing the capabilities of traditional RBAC models [27] [42].

OpenMRS supports interoperability with laboratory, pharmacy, and mHealth applications, utilizing international standards such as HL7 FHIR and dedicated integration modules to enhance data exchange and facilitate advanced clinical workflows [42]. Pharmacy integration efforts for OpenMRS are ongoing, with modules proposed and developed for medication dispensing,

inventory management, and role-restricted access management. While international case studies document the potential of such modules, widespread real-world use remains in progress; challenges include ensuring compliance with local data privacy regulations and embedding granular patient-consent functions [27].

2.2.3 Commercial EHR Platforms: SeriousMD and iClinicsSys

In the private healthcare sector, commercial EHRs such as SeriousMD and iClinicSys are widely utilized for patient management, caregiver logins, and real-time health record review. SeriousMD is a cloud-based platform that offers patient and caregiver interfaces; however, technical reviews often highlight unspecified or undocumented privacy and access control protocols, leaving unanswered questions about the adequacy of consent-driven, pharmacist-specific access. iClinicSys, endorsed by the Department of Health, is recognized for its efficient patient tracking and is implemented in both public and private healthcare centers [43]. Nevertheless, as with other legacy and commercial systems, transparency remains limited regarding extensibility, external module addition, and the sophistication of privacy mechanisms, particularly with respect to RBAC, consent management, and pharmacist access [44].

Table 1. Comparison of the Related Systems.

Systems	Purpose	Privacy and Access Control Focus	Technology Use	Key Features	Learning Focus
CHITS	Support primary healthcare data collection and public health reporting in LGUs	Basic access control with limited advanced privacy mechanisms	Open-source system with limited interoperability	Patient registration, primary care records, and public health program monitoring	Focus on operational efficiency in public healthcare
OpenMRS	Provide a flexible and extensible EHR platform for diverse clinical settings	Advanced RBAC and ReBAC through extensions	Open-source, modular architecture supporting HL7 FHIR	Modular plug-ins, extensible workflows, interoperability support	Emphasis on system extensibility and interoperability
SeriousMD / iClinicSys	Manage patient records and clinic workflows in public and private healthcare facilities	Privacy mechanisms not fully disclosed	Proprietary, cloud-based platforms	Patient management, appointment scheduling, and cloud-based record access	Focus on usability and clinical efficiency

RxGuard	Secure management of electronic health records with controlled access for doctors, pharmacies, and administrators	Privacy-by-design with strict RBAC, consent enforcement, and administrative approval	Secure authentication, encryption, audit logging, and modular system design	Role-based dashboards, consent-driven patient records, admin verification, and pharmacy validation	Emphasis on information security, data privacy, ethical compliance, and access control design

CHAPTER III
FRAMEWORK AND METHODOLOGY

This chapter discusses the framework and methodology used in the development of RxGuard. The chapter explains how the system operates through its conceptual framework and describes the process undertaken in developing the system.

3.1 CONCEPTUAL FRAMEWORK

The conceptual framework of RxGuard illustrates how the system transforms traditional paper-based medical records into a secure and accessible electronic health record platform. The framework follows the **Input–Process–Output (IPO)** model to clearly describe how the system functions.

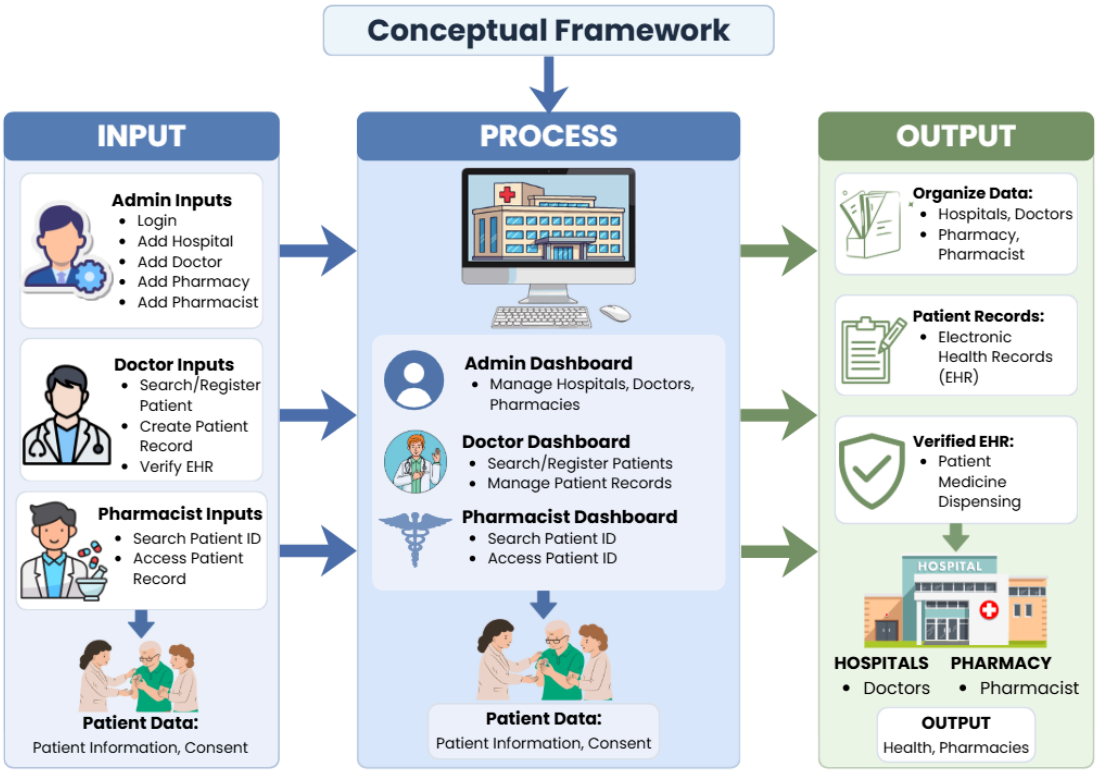

Figure 1. Conceptual Framework of RxGuard

The system's input consists of raw medical records submitted in various formats, including scanned documents, handwritten notes, and laboratory reports. These records serve as the primary source of patient information and may come from hospitals and clinics. At this stage, the data is unstructured and may contain inconsistencies, noise, or duplicates. The system accepts these inputs as digital images or files, which are then prepared for further processing. Proper data capture at the input stage is important to ensure accuracy and reliability in the succeeding steps of the system.

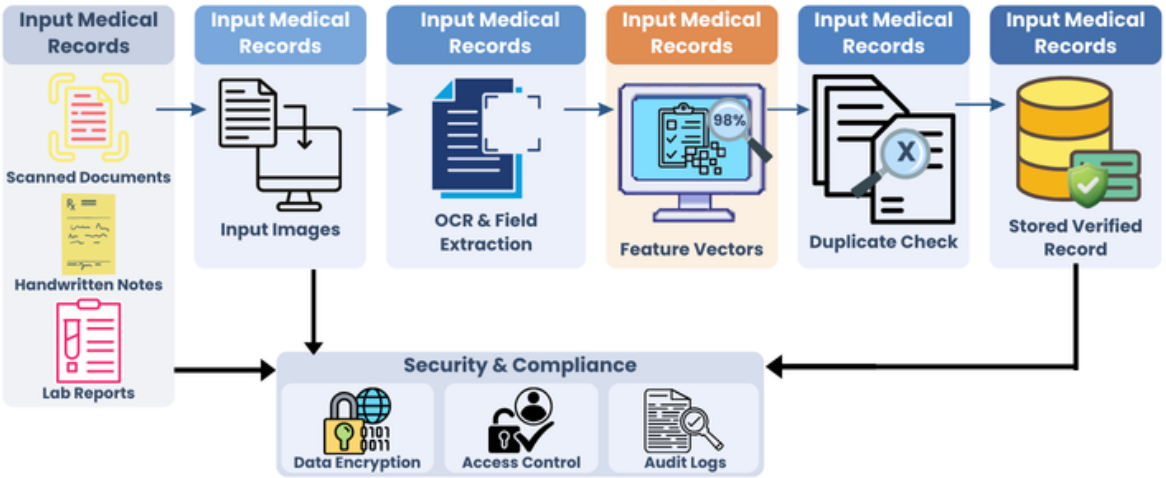

Figure 2. Process Flow of the Study

The process component represents the core operations performed by the RxGuard system. Upon login, users undergo authentication and are assigned system roles that determine their level of access. Patient data is then processed through registration and management modules, where consent and access control mechanisms ensure compliance with data privacy requirements. Medical documents are digitized using Optical Character Recognition (OCR) technology, and image deduplication is applied to prevent the creation of redundant or inconsistent records. Administrative verification further validates the authenticity of patient records before they are

securely stored in the electronic health record database. Prescription verification is then conducted to ensure that only authorized and accurate medical information is accessed by pharmacies.

The system's output includes digitized and verified electronic health records, authorized and role-based access to patient data, validated prescriptions, and secure medication dispensing. These outputs contribute to improved healthcare service delivery by minimizing record duplication, enhancing data integrity, and ensuring patient safety.

The conceptual framework provides a high-level representation of how RxGuard transforms healthcare data inputs into reliable and secure outputs. The detailed operational logic is further illustrated in the system flowchart.

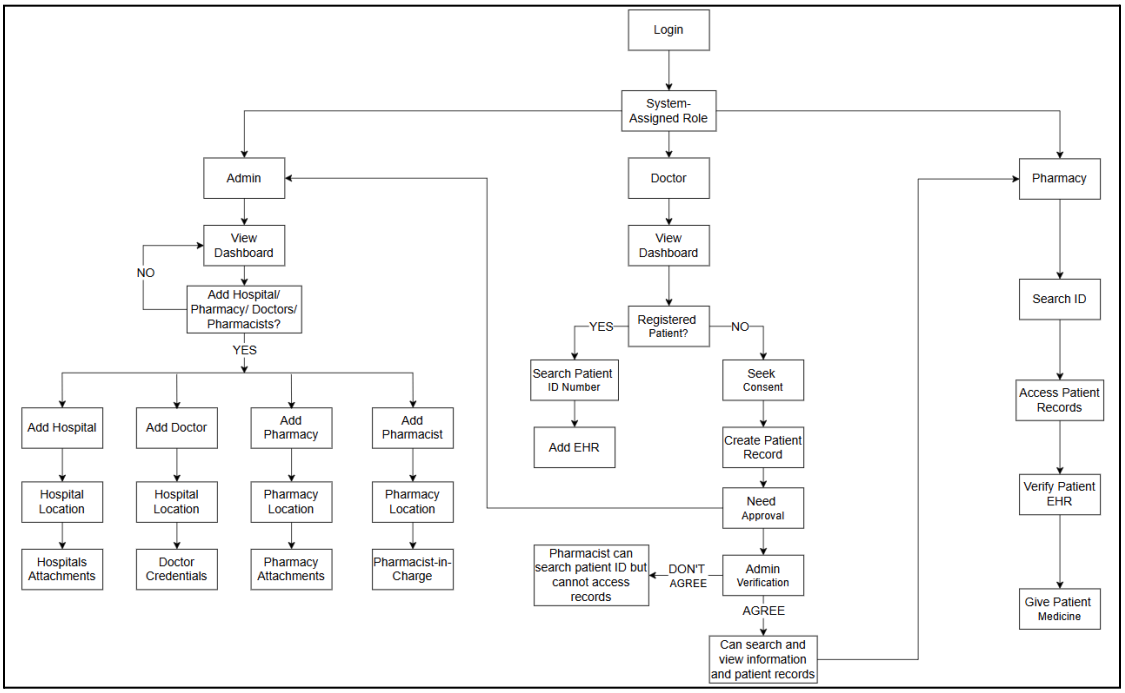

Figure 3. Flowchart of RxGuard

Figure 3 illustrates the system flow of RxGuard, beginning with user authentication and system-assigned role identification. Upon successful login, users are directed to role-specific functions, such as those of administrators, doctors, or pharmacy personnel. Administrators manage institutional records, including those of hospitals, doctors, pharmacies, and pharmacists, and are responsible for verifying patient records before granting access to them. Doctors handle patient registration, seek consent when necessary, and create or update electronic health records, which require administrative approval before becoming accessible. Pharmacy users are limited to searching for patient identification numbers, accessing verified records, and validating prescriptions before dispensing medication. Throughout the process, role-based access control, consent validation, and administrative verification are enforced to ensure data integrity, security, and compliance with healthcare data protection standards.

3.2 METHODOLOGY

This study adopts the Agile Scrum Methodology in the development of RxGuard, a Computer Vision-Based Electronic Health Record (EHR) system for medical record digitization with image deduplication. Scrum is an iterative and incremental software development framework that emphasizes flexibility, continuous feedback, and close collaboration among stakeholders. This methodology is appropriate for GuardRx as the system requires frequent refinement to meet healthcare workflow requirements, data privacy regulations, and cybersecurity standards.

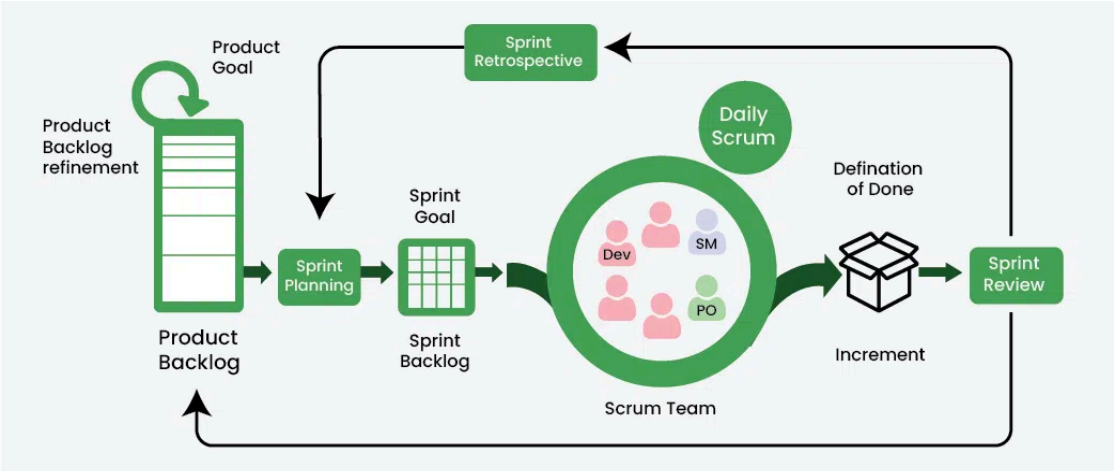

Figure 4. Agile Scrum Methodology

As illustrated in Figure 4, the development process of RxGuard will be divided into seven key phases: (a) product backlog and product goal, (b) sprint planning, (c) sprint development, (d) daily scrum, (e) sprint review, (f) sprint retrospectives, and (g) increment and definition of done during implementation. Each phase will steer the systematic design, development, and evaluation of the system, ensuring alignment with both instructional and technological objectives.

a. Product Backlog and Product Goal

The Scrum process begins with defining the product goal, which for the RxGuard system is the development of a secure and reliable medical record platform that utilizes computer vision to extract, validate, and deduplicate medical documents. The product backlog contains prioritized user stories and technical tasks related to document ingestion, OCR accuracy, image deduplication, and data security. Backlog refinement is conducted regularly to incorporate feedback from testing, emerging technical requirements, and performance results from previous sprints. Tasks related to improving image preprocessing, refining OCR models, and tuning similarity thresholds for duplicate detection are continuously updated to ensure that the system evolves toward higher accuracy and reliability.

b. Sprint Planning

During Sprint Planning, the Scrum team selects backlog items that can be realistically completed within the sprint, considering team capacity and technical complexity. For RxGuard, sprint goals are often centered on enhancing specific components of the computer vision pipeline, such as improving preprocessing quality, increasing OCR accuracy, or reducing false positives in image deduplication. The selected backlog items are transformed into a sprint backlog with clearly defined deliverables, including preprocessing modules, trained models, or validated deduplication thresholds. Technical decisions, such as the choice of OCR engine, feature extraction method, and similarity metrics, are finalized at this stage. This ensures that development efforts are focused, measurable, and aligned with the overall system objectives.

c. Sprint Development

Sprint Development is the core execution phase where the RxGuard system's computer vision and image deduplication functionalities are implemented and refined. The process begins with data preprocessing, where scanned and photographed medical documents undergo normalization techniques such as deskewing, perspective correction, contrast enhancement, adaptive thresholding, and denoising using OpenCV and morphological operations. To improve model robustness and generalization, image augmentation techniques, including rotation, blur, brightness variation, and simulated stains or folds, are applied during training. OCR and information extraction follow a two-stage approach, where the OCR layer converts document images into text using engines such as Tesseract with layout preprocessing or transformer-based OCR models. The document understanding layer then extracts structured fields, such as patient names, medication names, dosages, dates, and identification numbers, using layout-aware transformers like LayoutLMv3 or Donut-style models. Postprocessing techniques, including

medical dictionary normalization, fuzzy string matching, and rule-based corrections, are applied to correct formatting inconsistencies and enhance data accuracy. Image deduplication is implemented through a combined pipeline that first applies perceptual hashing methods for fast detection of exact or near-exact duplicates, followed by vector embedding comparison using CNN or vision transformer embeddings with cosine similarity for more precise validation. OCR-content hashing is optionally integrated to detect duplicates based on extracted textual content in addition to visual similarity. Similarity thresholds are empirically tuned using validation datasets to strike a balance between sensitivity and specificity.

d. Daily Scrum

The Daily Scrum is conducted to synchronize the team's progress and identify blockers related to model performance, data quality, or system integration issues. Team members discuss completed tasks, ongoing preprocessing or model training activities, and challenges such as noisy inputs or misclassified duplicates. Issues discovered during OCR extraction or similarity checking are raised promptly to enable timely adjustments. This daily coordination supports continuous improvement, ensuring that sprint goals remain achievable.

e. Sprint Review

In the Sprint Review, the team demonstrates the completed increment, which may include improved preprocessing outputs, higher OCR extraction accuracy, or enhanced duplicate detection performance. Stakeholders are shown how medical documents are ingested, processed, deduplicated, and stored as verified records. Feedback is collected regarding system usability, accuracy, and reliability, particularly in handling real-world document variations. The reviewed increment serves as a validated step toward the product goal and informs backlog refinement.

f. Sprint Retrospective

The Sprint Retrospective allows the team to reflect on technical and process-related aspects of the sprint. Discussions focus on what worked well in the computer vision pipeline, what caused errors or delays, and how preprocessing, OCR, or deduplication strategies can be improved. Lessons learned such as the need for better augmentation strategies or improved similarity threshold tuning, are documented. These insights are translated into actionable improvements for future sprints, reinforcing continuous learning and system enhancement.

g. Increment and Definition of Done

Each sprint produces a working increment that meets the defined acceptance criteria, known as the Definition of Done. For RxGuard, this includes accurate extraction of medical information, reliable duplicate detection, secure storage of verified records, and complete provenance metadata such as source image identifiers and confidence scores. Maintaining provenance and confidence information enables human-in-the-loop review, supporting system transparency. Through iterative increments, the Scrum framework ensures that the RxGuard system steadily advances toward a robust, secure, and production-ready platform for managing medical records.

Data Privacy, Ethics, and Regulatory Compliance

The proposed system is designed with data privacy and information security as core principles, recognizing that medical records constitute highly sensitive personal and health information. In compliance with applicable data protection regulations, such as the Data Privacy Act of 2012 (RA 10173) and generally accepted international standards for healthcare data protection, the system incorporates technical, administrative, and organizational safeguards to protect patient data throughout its lifecycle.

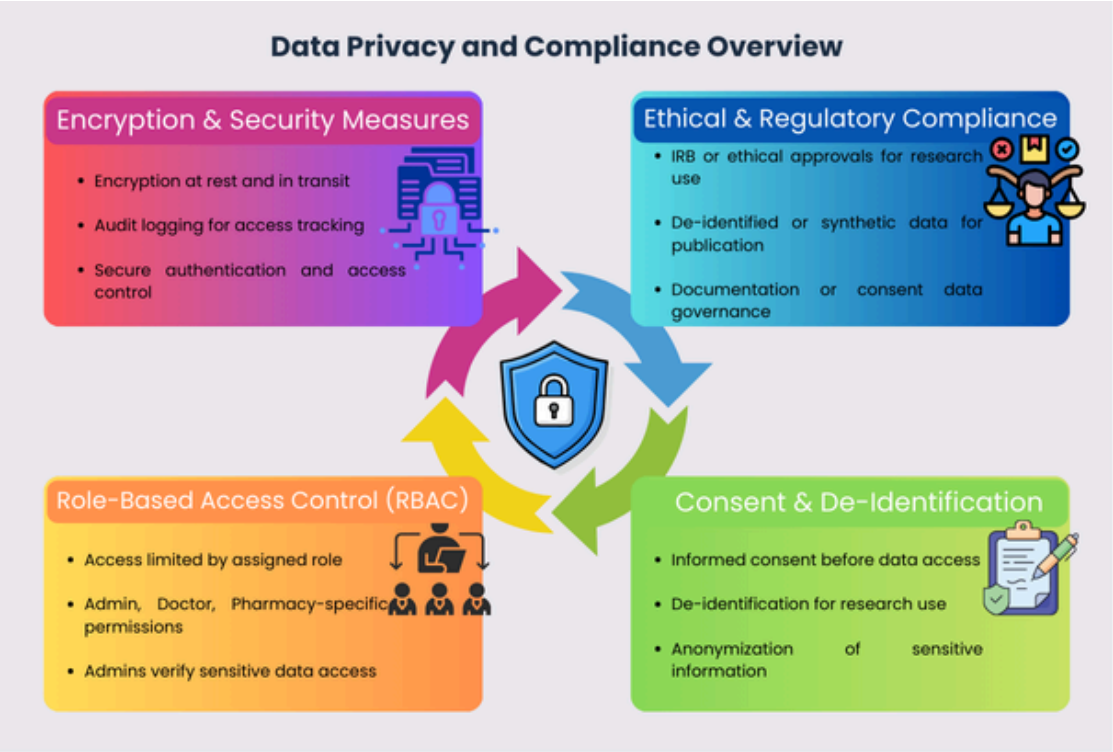

Figure 5. Data Privacy, Ethics, and Regulatory Compliance

a. Encryption and Security Measures

The system implements strong encryption mechanisms to protect sensitive medical data both at rest and in transit. All patient records, credentials, and institutional data are stored in encrypted form to prevent unauthorized disclosure in the event of a data breach. Secure communication protocols are used to ensure that data exchanged between users and the system cannot be intercepted or altered. Additionally, audit logging is enforced to record access and modification activities, supporting accountability, monitoring, and incident investigation.

b. Role-Based Access Control (RBAC)

Role-Based Access Control is enforced to ensure that users can only access data necessary for their assigned responsibilities. Administrators,

doctors, and pharmacy personnel are granted distinct permissions based on their roles within the healthcare workflow. This approach adheres to the principle of least privilege, minimizing the risk of data misuse or accidental exposure. Administrative verification is required before sensitive patient records become accessible, ensuring controlled and justified access to medical information.

c. Consent and De-Identification

The system prioritizes patient autonomy by requiring informed consent before creating or sharing medical records. When a patient is not yet registered, doctors must obtain explicit consent before recording personal and health-related data. For secondary use cases such as research, analytics, or system testing, patient data is de-identified or anonymized to remove personally identifiable information. This ensures that privacy is preserved while still allowing meaningful system evaluation and academic use.

d. Ethical and Regulatory Compliance

The system is designed to comply with data protection laws, such as the Data Privacy Act of 2012 (RA 10173), and recognized healthcare data governance standards. Any use of real patient data for research or publication purposes requires prior ethical approval or clearance from the Institutional Review Board (IRB). To further reduce privacy risks, de-identified or synthetic data is used whenever possible. Clear documentation of consent, access policies, and data governance procedures is maintained to ensure transparency, accountability, and compliance with relevant regulations

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PANEL COMMENTS IN OUR OUTLINE DEFENSE:

PARTICULARS	COMMENTS/SUGGESTIONS
Chapter 1: Introduction	- Consider the form, are you going to standardize the field - Handwriting of doctors to the OCR, look for OCR transforming text to its digital form → Evaluation (WER) - Check the actual process, especially in clinics - Strengthen the digitalization not solely for paperless - (Technical Scope Supposedly) Refs: Scope + Limition →improve refs Ma'am Ivy
Chapter 2: Review of Related Literature & Theoretical Framework	- Deduplication; there are techniques (check and validate) → do have AI?, approach with AI include - Search for St. Luke's Hospital - Encryption Method - Authentication
Chapter 3: Methodology	- Strengthen Data Privacy and Compliance (?) Overview
Other Comments/Remarks	- Objectives (UPDATE) - Replace (1)

	- Check algorithms (many) (2) - Replace (4) - Consider Next Visit Scenario (append) - Consider the context (2nd g / transfer hospital) - Scanning of docs for 1st phase, the 2nd is input - Encryption Method Narration (?) - Authentication Narration(?) - Refer to the manuscript of the panel members - Check the style sheet - Figure must be readable - Design Data Flow (Database Design) - ERD and Data Dictionary - Data Flow Diagram (DFD) and Sequence Diagram - Use Case - Prototypes